WORKED SOLUTION

STEP II 2019, Question 2

Pure • Integrals of Inverse Functions

STEP II 2019, Question 2

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

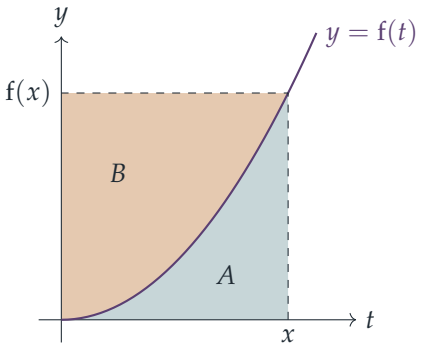
The key idea

The graph of $y = \mathrm{f}(t)$ cuts the rectangle with corners $(0,0)$ and $(x, \mathrm{f}(x))$ into two pieces. One piece is the area under the curve, and the other is the area between the curve and the y-axis, which is the integral of the inverse function. So the two integrals add up to the area of the rectangle. Parts (i) and (ii) define functions only implicitly, so we can't integrate them directly, but their *inverses* are simple cubics. The identity lets us swap a hard integral for an easy one.

Solution

The opening result

Since $\mathrm{f}'(t) > 0$ for $t > 0$ and $\mathrm{f}(0) = 0$, f is strictly increasing on $[0, x]$, so $\mathrm{f}(t) > 0$ for $t > 0$ and f has an inverse f^{-1} from $[0, \mathrm{f}(x)]$ to $[0, x]$. The same curve is described by $y = \mathrm{f}(t)$ and by $t = \mathrm{f}^{-1}(y)$.



The region A, between the curve and the t-axis for $0 \leq t \leq x$, has area

$$A = \int_0^x \mathrm{f}(t)\,\mathrm{d}t.$$

The region B, between the curve and the y-axis for $0 \leq y \leq \mathrm{f}(x)$, is made of horizontal strips. The strip at height y runs from $t = 0$ to $t = \mathrm{f}^{-1}(y)$, so

$$B = \int_0^{\mathrm{f}(x)} \mathrm{f}^{-1}(y)\,\mathrm{d}y.$$

Because the curve is increasing and goes from $(0,0)$ to $(x, \mathrm{f}(x))$, A and B fit together exactly, with no overlap, to fill the rectangle $0 \leq t \leq x$, $0 \leq y \leq \mathrm{f}(x)$. That rectangle has area $x\,\mathrm{f}(x)$, so

$$\int_0^x \mathrm{f}(t)\,\mathrm{d}t + \int_0^{\mathrm{f}(x)} \mathrm{f}^{-1}(y)\,\mathrm{d}y = x\,\mathrm{f}(x),$$

as required.

Part (i)

Write $\phi(y) = y^3 + y$. Then $\phi'(y) = 3y^2 + 1 > 0$, so ϕ is strictly increasing, and $\phi(y) \to \pm\infty$ as $y \to \pm\infty$. So for each real t there is exactly one real y with $\phi(y) = t$, and this is $\mathrm{g}(t)$. In other words, $\mathrm{g} = \phi^{-1}$.

$\mathrm{g}(0) = 0$. Putting $t = 0$ gives $\mathrm{g}(0)(\mathrm{g}(0)^2 + 1) = 0$. Since $\mathrm{g}(0)^2 + 1 \geq 1$ for real $\mathrm{g}(0)$, it follows that $\mathrm{g}(0) = 0$.

$\mathrm{g}'(t) > 0$. Differentiating $\mathrm{g}(t)^3 + \mathrm{g}(t) = t$ implicitly with respect to t,

$$(3\,\mathrm{g}(t)^2 + 1)\,\mathrm{g}'(t) = 1 \quad \Longrightarrow \quad \mathrm{g}'(t) = \frac{1}{3\,\mathrm{g}(t)^2 + 1} > 0,$$

since the denominator is at least 1.

? **Ask yourself** We can't write $\mathrm{g}(t)$ down, so how can we integrate it?

We don't need to. g satisfies the conditions of the opening result, and its inverse is $\phi(y) = y^3 + y$, which is easy to integrate. Applying the result with $x = 2$ needs $\mathrm{g}(2)$, the real root of $y^3 + y = 2$. This root is $y = 1$, and it is the only one because ϕ is strictly increasing. So

$$\int_0^2 \mathrm{g}(t)\,\mathrm{d}t = 2\,\mathrm{g}(2) - \int_0^1 (y^3 + y)\,\mathrm{d}y = 2 - \left(\tfrac{1}{4} + \tfrac{1}{2}\right) = \frac{5}{4}.$$

Part (ii)

Comparing the two definitions, $\mathrm{h}(t)$ is the real solution of $y^3 + y = t + 2$, so $\mathrm{h}(t) = \mathrm{g}(t + 2)$. Substituting $u = t + 2$,

$$\int_0^8 \mathrm{h}(t)\,\mathrm{d}t = \int_2^{10} \mathrm{g}(u)\,\mathrm{d}u = \int_0^{10} \mathrm{g}(u)\,\mathrm{d}u - \int_0^2 \mathrm{g}(u)\,\mathrm{d}u.$$

For the first integral we use the opening result with $x = 10$. Since $2^3 + 2 = 10$, we have $\mathrm{g}(10) = 2$, and so

$$\int_0^{10} \mathrm{g}(u)\,\mathrm{d}u = 10 \times 2 - \int_0^2 (y^3 + y)\,\mathrm{d}y = 20 - (4 + 2) = 14.$$

Using part (i) for the second integral,

$$\int_0^8 \mathrm{h}(t)\,\mathrm{d}t = 14 - \frac{5}{4} = \frac{51}{4}.$$

Another way. We can also apply the picture to h directly. This time $\mathrm{h}(0) = 1$, not 0. The curve runs from $(0,1)$ to $(8,2)$, and $\mathrm{h}^{-1}(y) = y^3 + y - 2$. The area under the curve is the 8×2 rectangle minus the 0×1 rectangle (which is empty here) minus the area to the left of the curve for $1 \le y \le 2$:

$$16 - \int_1^2 (y^3 + y - 2)\,\mathrm{d}y = 16 - \left(\tfrac{15}{4} + \tfrac{3}{2} - 2\right) = 16 - \tfrac{13}{4} = \tfrac{51}{4}.$$

Checks

- $0 \le \mathrm{g}(t) \le 1$ on $[0,2]$, so the integral in (i) must lie between 0 and 2. Also, g is concave there (it is the inverse of an increasing convex function), so the integral is at least the triangle area $\frac{1}{2} \cdot 2 \cdot 1 = 1$. The answer $\frac{5}{4}$ fits.
- In (ii), $1 \le \mathrm{h}(t) \le 2$ on $[0,8]$, so the integral lies between 8 and 16, and 12.75 does too.
- Solving the cubics numerically at many points and integrating by Simpson's rule gives 1.25000 and 12.75000, which agree with the exact answers.

Common slips

- Using the wrong upper limit for the inverse integral: it is $\mathrm{g}(2) = 1$, not 2.
- Applying the opening result to h as if $\mathrm{h}(0) = 0$. Since $\mathrm{h}(0) = 1$, the picture has an extra strip, and the identity needs adjusting (or use $\mathrm{h}(t) = \mathrm{g}(t+2)$).
- Forgetting to justify that g is well defined, that is, that $y^3 + y = t$ has exactly one real root.

Beyond STEP

The opening identity is the equality case of *Young's inequality*: for increasing f with $\mathrm{f}(0) = 0$ and any $a, b \ge 0$, $ab \le \int_0^a \mathrm{f} + \int_0^b \mathrm{f}^{-1}$. Taking $\mathrm{f}(t) = t^{p-1}$ gives $ab \le a^p/p + b^q/q$, the starting point for Hölder's inequality in analysis. The fact that g is differentiable with $\mathrm{g}' = 1/\phi'(\mathrm{g})$ is the *inverse function theorem*. Abbott's *Understanding Analysis* proves this carefully.

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